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Subject: Data Analytics
Research Assignment 1

1. A Database - is a system used to collect, organized and store information in a way that it make it easy to search update and manage.

Make it like is a highly intelligent filing cabinet, instead of shuffling through paper folders, you can instantly retrieve exactly what you need, filter, sort it, and connect related pieces of information together.

- Database are essential in the modern life world because almost every organization depends on data to operate efficiently. They store huge amount of data information securely, reduce duplication, improve accuracy and allow multiple users to access data at the same time.

Without database, managing record manually would be slow, confusing, and error, prone.

Real world example

Banking system

Banks use databases to store customer information, account balances, transaction history, and loan details. Whenever you withdraw money from an ATM, or transfer funds online, the database updates the information instantly.

Hospital Management system

Hospitals use database to keep patients records, doctor schedules, medical history and billing information.

This helps doctors and staff quickly access accurate patient data and provide better health care services.

2. Data warehouse - is a large storage system that collects and stores data from different sources within an organization for reporting, analysis and decision-making.

It is designed to help businesses study trends, compare performance, and make strategic decisions using historical data.

A regular database is mainly used for daily operations such as capturing, updating and managing current information. For example, a bank database records deposits and withdrawals every second.

A data warehouse, on the other hand, stores large amounts of past and current data so that managers and analysts can study patterns over time.

Regular Database	Data warehouse
Use for daily operations	used for analysis and reports
Stores current and detailed data	Store historical and summarized data
Frequently updated	updated periodically
Supports transaction like sales and payment	Support business intelligence and decision making
Faster for simple transaction	Faster for complex queries and reports

Why would a company use a data warehouse

1. Better business decision-making
A data warehouse gives executives and analysts a single source of truth.
2. Identifying trends overtime
It stores years of historical data, companies can spot patterns, seasonal sale spikes, customer behavior shifts, or cost trends.
3. Faster and Smarter reporting
Running complex reports on a regular database slows down daily operations. A data warehouse handles heavy analytical queries separately, keeping both systems running smoothly.
4. Connecting data from multiple systems
Large companies use dozens of different software tools. A data warehouse brings all that scattered data together, eliminating silos and giving a complete picture of the business.

3. Data lake - is a storage system that keeps large amount of raw data in its original format until it is needed. The data can be structured, semi structured or unstructured, such as text files, Videos, emails, social media posts, images and sensor data.

The main purpose of data lake is to allow organization to store all types of data in one place for future analysis, machine learning, and big data processing.

If data warehouse is a clean, organized library where every book is catalogued and shelved. A data lake is a giant reservoir where all water (data) flows in freely and is only filtered used when needed.

Data lake	Data warehouse
Store raw and unprocessed data	Store processed and organized data
Can handle structured and unstructured data	Mainly handles structured data
Flexible and low-cost storage	More structured and organized
Used by data scientists and analysts	Used by managers and business users
Data is store before processing	Data is processed before storage

When would a business prefer a data lake?

1. Artificial intelligence & machine learning

AI models require huge volumes of diverse data to train on images & audio.

2. Real time & IoT Data

Devices like smart sensors, GPS trackers, and wearables generate continuous streams of data. A data lake can ingest this instantly without waiting for transformation.

3. Diverse data formats

If a business collects social media comments, call recordings, images and emails, data lake handles all formats in one place, unlike a warehouse which only accepts structured data.

4. Data Mart - is a smaller, focused subset of a data warehouse that is tailored to meet the specific needs of a particular department or business unit. Instead of giving everyone access to the entire warehouse, a data mart delivers only relevant slice of data to the people who need it.

A data mart is usually created from a larger data warehouse. While the data warehouse stores information for the whole organization, the data mart stores selected data for one department.

For example

- * The main data warehouse may contain all company information
- * The sales data mart will only contain sales-related data.
- * Build from the data warehouse (dependent data mart)
- * Focused on one subject area or department
- * Smaller and faster to query than the full warehouse
- * Easier to manage with tighter security per department

Who typically uses data Mart?

Department	What they use it for?
Sales team	Tracking revenue, targets and pipeline performance
Marketing team	Comparing performance, customer segmentation
Finance team	Budgets, forecasts, expenses, tracking
HR Department	Employee performance, headcount, recruitment metrics.

5. Structured data - is highly organized and stored in a fixed format, usually in rows and columns like a table. Every piece of data has a clear label, type and place. It is the easiest type of data to store and analyze.

Example A student database containing student names, student numbers, marks and course details

Semi-structured data - does not follow a strict table format, but it still contains labels or tags that help organize the information. It is flexible fields can vary between records, and new fields can be added without breaking the system.

example Semi structured is like a CV, everyone includes their names and contact details,
Use emails or JSON/XML files where information is grouped using tags and headings

Unstructured data - has no fixed format or organization. It is more difficult to process and analyze because it contains free-form information.

It is the rawest form of data making up over 80% of all data generated in the world today. It cannot be stored in traditional tables and requires specialized tools like AI, machine learning or big data platforms to process.

example Videos, photos, audio recordings and social media posts

6. Data modeling - is the process of creating a visual representation of how data will be stored, connected and managed in a system. It helps designers understand the relationship between different type of data before building a database or information system.

Data modeling acts like a blueprint for a database system because it shows how information flows and how different data element relate to each other.

Modeling is important because it provides clarity as it provides a clear blueprint for developers, analyst and stakeholders to understand the data requirements.

It is efficiency, this will optimize database design, reducing redundancy and improving performance.

Consistency ensures data integrity by enforcing rules and relationships.

Communication acts as a common language between technical and non-technical teams.

Scalability makes future system expansion easier by providing a structured foundation.

1. Conceptual model - high level overview of data and relationships, focusing on business requirements rather than technical details. Focuses on entities (Customer, order, product) and the relationship between them. No attributes types, no keys, no implementation details.
e.g.

2. Logical Data model - more detailed, defining attributes, relationships and constraints without considering physical storage.
e.g. Used by database designers to plan structure before implementation.

7. Data mining - is the process of finding useful patterns, trends, and information from large amount of data.
Business use data mining to better understand customer, improve decisions and predict trends.

Data mining is used in business to understand customer behavior, improve marketing strategies, detect fraud, predict sales trends and recommend products to customers.

Two real-world examples

1. Amazon - recommendation engine

Amazon mines the purchase history, browsing behavior, wish lists and ratings of hundred of millions of customers. Using collaborative filtering and association rule mining.

2. Bank Fraud detection

Bank use data mining to detect unusual transaction that may indicate fraud or stolen bank cards.

3. Data management system - is a software that enables users and applications to create, read, update, delete and manage data in a structured and secure way.

MySQL - A popular open-source relational database that is fast and easy to use.
websites and small-to small medium business application.

PostgreSQL - An advance open source database known for reliability and powerful features.
Banking systems and data analytics applications.

Oracle database - A commercial DBMS designed for large organisation and enterprise systems.
Large companies handling secure financial or business data.

MongoDB - A NoSQL database that stores flexible document-based data instead of tables.
Social media apps and applications with unstructured data.

9. ETL - is data integration process that moves data from one or more sources systems into a target destination.
It moves data to data warehouse or analytics platforms in a clean, structured and usable form.
It is the backbone of modern pipelines and intelligence infrastructure.

1. Extract - the extract step involves collecting data from different sources such as database (MySQL, Oracle), flat files (CSV, Excel), cloud platforms (Google Analytics), Streaming Source (Clickstreams)

2. Transform - It involves cleaning, organizing and converting the data into a useful format.
This may include removing errors, fixing missing information, combining data and changing format.

3. Load - stores the cleaned data into a database or data warehouse where it can be analyzed.
Common target destination
Data warehouse
Data lakes
Analytical database
BI dashboards

ETL is important because it improves data quality, combines data from different sources, make data easier to analyze, helps businesses make accurate decisions and it saves time by automating data preparation.

10. CSV (Comma-Separated Value)

A CSV file store data in plain text using rows and columns separated by commas

Features

Simple and easy to read, works with spread sheet and databases & light weight format

Used for small datasets, spreadsheets and simple data sharing

2. JSON (JavaScript object Notation)

JSON stores data in a flexible text format using Key-Value pairs

Features

easy for web applications to use, support nested and semi-structured data and human readable data
Best used for APIs, web applications, semi-structured

3. Parquet

Parquet is a column-based storage format designed for big data processing

Features
Files are highly compressed, very fast for analytics queries and saves storage space

Used for big data analytics and cloud data processing

4. Avro - is a compact data storage format commonly used in big data systems. Support data schemas, good for data streaming. Used data streaming and bigw

11. Artificial intelligence - is the ability of computers or machines to perform task that normally require human intelligence, such as learning, problem-solving, understanding language, and making decisions.

Types of AI

Narrow AI (Weak AI)

Narrow AI is designed to perform one specific task only
~~example it is designed to perform one specific task~~
only
google maps uses AI to give directions and traffic updates.

General AI (Strong AI)

is a type of AI that can think and learn like a human across many task
eg A human-like robot that can learn different skills and make decisions independently

12. Machine Learning is a branch of AI where computers learn from data and improve their performance without being directly programmed for every task

Difference between ML and TP

Traditional Programming	Machine Learning
Programmer writes all rules	System learns pattern from data
Fixed instructions	Learn and improves over time
Best for simple tasks	Best for predictions and pattern recognition.

Types of machine learning

1. Supervised learning
- The computer learns using labeled data such as email spam detection

2. Unsupervised Learning
the computer finds patterns in unlabeled data
e.g. grouping customers by shopping behavior

3. Reinforcement learning
the computer learns through rewards and penalties
e.g. Robots learning how to walk or play games

13. What is deep learning
deep learning is more advanced type of machine learning
that uses systems called neural networks to learn
from large amount of data

The relationship is that AI is the broad field, Machine learning is a part of AI and deep learning is specialized part of machine learning.

Neural networks - are computer systems inspired by the human brain. They contain layers of connected nodes that process information and learn patterns from data.

How neural networks work

- Data enters the network
- The system processes the data through layers
- The network learns patterns by adjusting connections
- It produces prediction or decision.

14. NLP - is a field of AI that helps computers understand, interpret, and respond to human language

Real-world Application of NLP

1. Voice assistants - Siri and Google Assistant understand voice commands

2. Translation Apps

Google Translate, translate language instantly

3. Chatbot - Customer service chatbots answer questions automatically on website.

15 LLM is an advanced AI model trained on massive amounts of text data to understand and generate human like language

Standard ML model	Large Language model
Trained for specific task	Handles many language tasks
Smaller datasets	Massive text datasets
Limited language understood	Advanced human-like response

example of LLMs

1. ChatGPT - used for answering questions, writing, coding and conversations
2. Gemini - used for research, writing and AI powered assistance

16 Generative AI - is AI that can create new content such as text, images, music, videos or code. It works as it learns patterns from large datasets and then generates new content based on what it learned

examples such as ChatGPT as it generates text, answers questions and helps with writing
DALL-E creates images from text descriptions
GitHub Copilot helps programmers with code

17. A training dataset is a collection of data used to teach an AI or machine learning model how to perform a task

Importance of good training data

- It improves AI accuracy
- It helps the model learn correctly
- It reduces errors and unfair results

Problems with Biased or incorrect data

- The AI may make wrong predictions
- It may treat some groups unfairly
- Results may become unreliable

18 Computer vision - is a field of AI that helps computers understand and interpret images and videos

How machines "See" Images

- Computers analyze images using algorithms that detect
- Shapes - Pattern - Faces
 - Colors - Objects

The AI then identifies and interprets what is in the image

Real world use of Computer vision

1. Facial recognition

Smartphones unlock using face detection technology

2. Self-driving cars

Autonomous vehicles use computer vision to detect road traffic signs and pedestrians.

API
19 ~~APP~~ is a tool that allows different software applications to communicate and share information with each other

Simple real-world analogy

An API is like a waiter in a restaurant

* You place an order with the waiter, the waiter takes the request to the kitchen, the kitchen prepares the food then the waiter brings the food back to you.

In the same way, an API takes request from one application, send them to another system and returns response

API help different apps work together, share data quickly, save development time and connect websites, app and cloud services

e.g. A weather app uses an API to get weather information from another service

20 API Key - is a secret code used to identify and authorize a user or application when accessing an API

Purpose of API Key

- It helps to control access to APIs, track usage
- Protect system from unauthorized users.

How it should be protected

API Key should be kept private, store securely in environment variables or secure system and never be shared publicly online.

Risks if exposed publicly

If someone steals an API Key they can misuse API. They may access private data and the owner may face security problem or extra costs.

21. A REST API is a type of API that follows rules for communication between systems over the internet

REST stand for
Representational State Transfer

REST APIs commonly use HTTP methods to perform actions

Main HTTP Methods		
method	Purpose	Example
Get	Retrieve data	Get user information
Post	Create new data	Create a new account
Put	Update existing data	Update profile details
Delete	Remove data	Delete user account

22 JSON is a lightweight text format used to store and exchange data between systems

Why JSON commonly used?

- easy for human to read, easy for computers to process & work well with APIs, web application.

23 Prompt engineering - is a skill of writing clear and effective instructions for AI tools like Chat GPT or Claude to get better response

Why it is important?

good prompts improve AI answers, save time, produce more accurate results and help AI understand the task clearly

example Poor prompt

Tell me about marketing

Improved prompt

Explain digital marketing in simple terms and give three examples.

24 MAP Function - is used in programming to apply the same action to every item in a list and return a new list with update values

Python example

Python

```
numbers = [1, 2, 3, 4]
```

```
Squared = List(map(lambda x: x*x, numbers))
```

```
Print (Squared)
```

output

[1, 4, 9, 16]

The example squares every number in the list

25 Cloud computing is the delivery of computing service such as storage, software and servers over the Internet instead of using local computers only

Main Cloud service models

model	Meaning	Description	example
IaaS	Infrastructure as Service	Provide virtual servers and storage	Amazon web Service
PaaS	Platform as a Service	Provide tools for building application	Google cloud
SaaS	Software as a Service	Provides software through the internet	microsoft 365

26 Version control System - helps developers track and manage changes made to files and code over time
What is Git?

is a popular version control system used by developers and data analysts to manage projects and collaborate with team

Why Git is widely used?

- Track changes in projects, allows teamwork, prevent loss of work and make it easy to restore previous versions

Important terms

Commit - is a saved snapshot of changes made to a project

Branch - is a separate version of a project used to test or develop new features without affecting the main project

27 A Jupyter Notebook - is an interactive tool used for writing and running code, displaying results and adding notes in one document

Common programming language

It is mostly common used with python why it is popular

Data analyst and data scientists use it because it allows code and explanations together, easy to visualize data, useful for testing and analysis and it supports charts and graphs

28 Python is a popular programming language known for being simple, powerful and easy to learn

why python is popular for data Analytics & AI

* easy syntax, large community support, many data and AI libraries, and works well with machine learning & automation

Common Python Libraries

Library

Purpose

Numpy

works with numbers and arrays

Pandas

Used for data analysis and tables

Matplotlib

Creates charts & graphs

Tensorflow

Used for artificial intelligence and deep learning

29 What is data privacy
Data privacy refers to protecting personal and sensitive information from unauthorized access, misuse or sharing

Data privacy is important because it protect personal information
Prevents identity theft and fraud, build trust between businesses and customers and also ensures information is used responsibly

General Data Protection Regulation (GDPR) is a law created by European Union to protect people's personal data and privacy

Countries where GDPR applies to
x Countries in the EU
x Any company worldwide that collects or processes data from EU citizens

Purpose of GDPR it gives people more control over their personal information and requires companies to ask permissions before collecting data, protect customer data properly and report data breaches

30. Data ethics - refers to using data responsibly, fairly and honestly
example of unethical use of data or AI

③ Selling personal data without permission
companies sharing customer information without consent
why it is a problem?
It violates privacy and trust

② Biased Hiring AI
An AI hiring system unfairly rejects certain groups of people
why it is a problem?
It causes discrimination and unfair treatment

③ Fake deepfake videos

Using AI to create fake videos of people saying things they never said
why it is problematic?
it spreads misinformation and damages reputations.

31. Data Governance - is the process of managing data properly to ensure it is accurate, secure, consistent and used responsibly across an organization.

Key component of good data governance.

Component	Description
Data quality	Ensuring data is accurate and complete
Data security	Protecting data from unauthorized access
Data policies	Rules for how data should be used
Data ownership	Defining who is responsible for data
Compliance	Following laws and regulation

Data governance is important because it improves decision-making, protects sensitive information, ensures legal compliance and maintains reliable data.

32 Bias in AI - happens when AI system produces unfair or inaccurate results because of biased training data or design

Real world example

Some facial recognition systems perform poorly on certain skin tones because they were trained using limited or unbalanced datasets

Possible consequences

- Unfair treatment
- Wrong decisions
- Discrimination
- Loss of trust in AI systems.

33 Differences between Data Analysts, Data Scientists, Data Engineering and AI engineers

Role	Main work	Skills	Common tools
Data Analyst	Analyze data and creates reports	Excel, SQL, Visualisation	Microsoft Excel, Power BI
Data scientist	Builds predictive models and analyze patterns	Statistics, ML, Python	Python, Jupyter notebook
Data engineering	Build systems for storing and moving data	Database, cloud systems	Apache Spark, Amazon web services
AI Engineer	Develops AI and machine learning systems	Deep learning, AI models	Tensorflow, Pytorch

34. Business Intelligence (BI) refers to technologies and tools used to analyze data and help businesses make informed decisions

Popular BI Tools

1. Power BI - used to create dashboards, reports and visualisations from business data

2. Tableau - used for interactive charts and data visualisation

BI tools help businesses to understand company performance track sales and profits, identify trends and make data-driven decisions.

35 One concept I found most interesting was GDPR because it made me realize how much personal information companies collect from people every day. I found it surprising that laws are needed to protect people's data and ensure companies use information responsibly. This topic helped me understand the importance of protecting personal data and being careful about what we share online.

36 One concept I found most difficult to understand was data Modeling because it required understanding how different data elements connect and relate to each other within a system. Initially found it challenging to identify relationships between tables and design data structures correctly. To improve my understanding:

37 I believe data and AI will play an important role in my future and everyday life by helping businesses make better decisions, improve efficiency, and automate tasks. In daily life, AI is already used in social media, online shopping, navigation and customer service. Learning about data and AI will help me adapt to future technology-driven workplaces and improve my career opportunities.

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